

Adversarial Attacks on Critical Infrastructure IDS: A Water Treatment Case Study

Cybersecurity Capstone Project

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Abstract

This project investigates the robustness of Machine Learning-based Intrusion Detection Systems (IDS) in Critical Infrastructure scenarios. Replicating the methodology proposed by Barletta and del Vescovo [1] for the Automotive domain, this work applies Black-Box Adversarial Machine Learning attacks to the Water Treatment domain using the CISS 2020 dataset. We evaluate Decision Tree, Random Forest, and XGBoost models against ZOO, Boundary, HopSkipJump, and Sign-OPT attacks. Results show that while ensemble models are robust against gradient-estimation attacks, they remain vulnerable to decision-based attacks like Sign-OPT. Notably, XGBoost demonstrates superior resilience against geometric attacks compared to Random Forest. Furthermore, we analyze the trade-off between feature engineering efficiency and adversarial vulnerability.

1 Introduction

The integration of Machine Learning (ML) in Cyber-Physical Systems (CPS) has improved anomaly detection capabilities. However, ML models introduce a new attack surface: Adversarial Examples. This project aims to verify the transferability of the attack pipeline defined in [1] from the *Automotive* domain (CAN bus data) to the *Water Treatment* domain (Sensor data), analyzing the impact on widely used classifiers.

2 Methodology

2.1 Dataset: CISS 2020 (SWaT)

The dataset used is the *Critical Infrastructure Security Showdown 2020* (CISS) [2], collected from the Secure Water Treatment (SWaT) testbed. Unlike the discrete CAN bus data used in the reference thesis, this dataset contains continuous physical measurements (e.g., water levels, flow rates).

2.2 Preprocessing and Leakage Removal

A critical phase of this project was the identification and removal of **Data Leakage**. The raw dataset contained metadata columns (e.g., **Attack ID**, **Intent**) that directly revealed the class label. To simulate a realistic IDS, these columns were removed, leaving only 58 physical sensor features. Subsequently, a **Feature Engineering** phase removed constant and highly correlated features (> 0.99), reducing the dimensionality to 46 features.

Feature	FIT101	LIT101	MV101	Label
Sample 1	0.896	0.652	1.0	0 (Normal)
Sample 2	0.000	0.647	0.5	0 (Normal)
Sample 3	0.507	0.420	0.5	1 (Attack)

Table 1: Example of refined dataset (Normalized 0-1).

2.3 Models and Attacks

We trained three models: **Decision Tree (DT)**, **Random Forest (RF)**, and **XGBoost (XGB)**. Using the *Adversarial Robustness Toolbox* (ART) [3], we executed four Black-Box evasion attacks:

- **ZOO**: Zeroth Order Optimization (Gradient estimation).
- **HopSkipJump**: Boundary-based attack (Geometric).
- **Boundary Attack**: Random walk optimization along the decision boundary.
- **Sign-OPT**: Sign-based gradient estimation (Adapted for binary classification).

3 Exploratory Data Analysis (EDA)

Before training the models, a comprehensive statistical analysis was conducted to understand the underlying data distribution and the physical relationships between the sensors of the Water Treatment plant.

3.1 Feature Correlation and Physical Constraints

The Correlation Matrix, shown in Figure 1, highlights strong linear relationships (Pearson coefficient > 0.98) between specific clusters of sensors. From a physical perspective, this redundancy is expected in a hydraulic process due to mass conservation laws and control logic (e.g., if a pump starts, the downstream flow rate must increase immediately).

However, from a Machine Learning perspective, this **multicollinearity** introduces redundancy without adding information gain. **Strategic Decision:** We performed a Feature Engineering step to remove these highly correlated features. This reduces the dimensionality of the search space for adversarial optimization algorithms (like ZOO), theoretically accelerating the generation of adversarial examples, as discussed in Section 6.

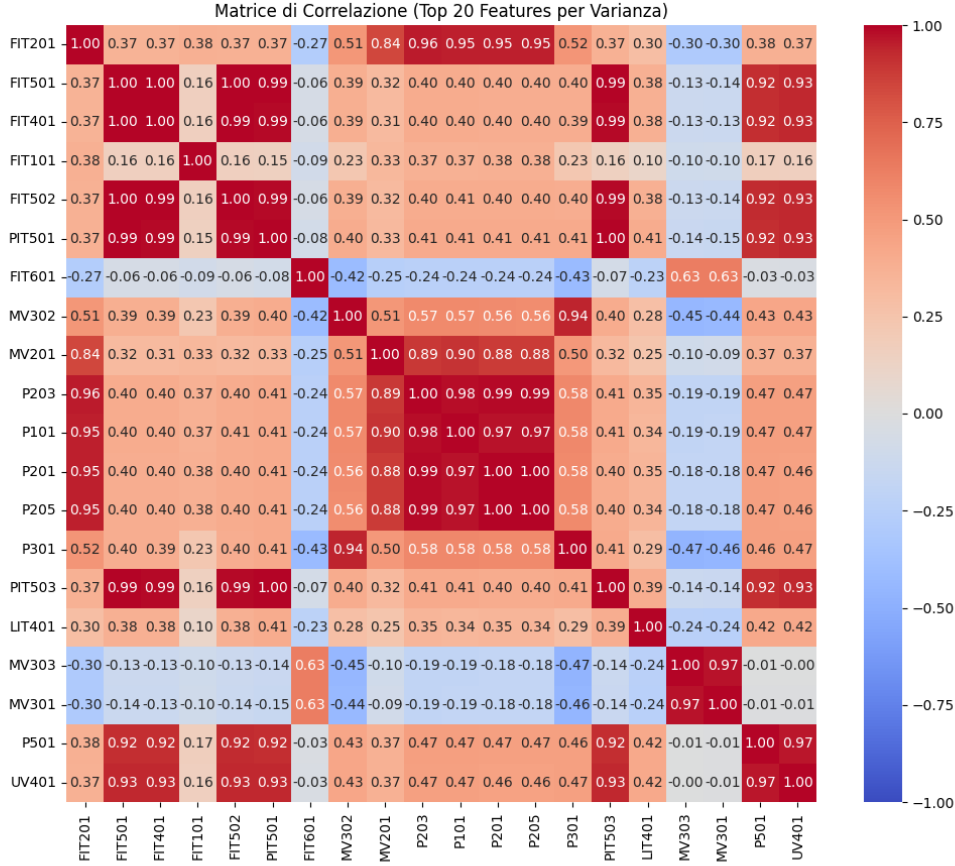


Figure 1: Correlation Matrix: Red blocks indicate redundant physical sensors (e.g., FIT501 vs UV401).

3.2 Class Distribution and Imbalance

The analysis of the target variable reveals a significant class imbalance, as shown in Figure 2. The dataset contains approximately 30,000 *Normal* instances against only 2,000 *Attack* instances (a ratio of roughly 15:1).

This imbalance reflects real-world Operational Technology (OT) environments, where cyber-attacks are rare events compared to normal operations. Consequently, relying solely on *Overall Accuracy* would be misleading (a dummy classifier predicting always "Normal" would achieve $\approx 94\%$ accuracy). Therefore, this study prioritizes **F1-Score (Weighted)** and **Recall** as the primary metrics to evaluate the IDS effectiveness.

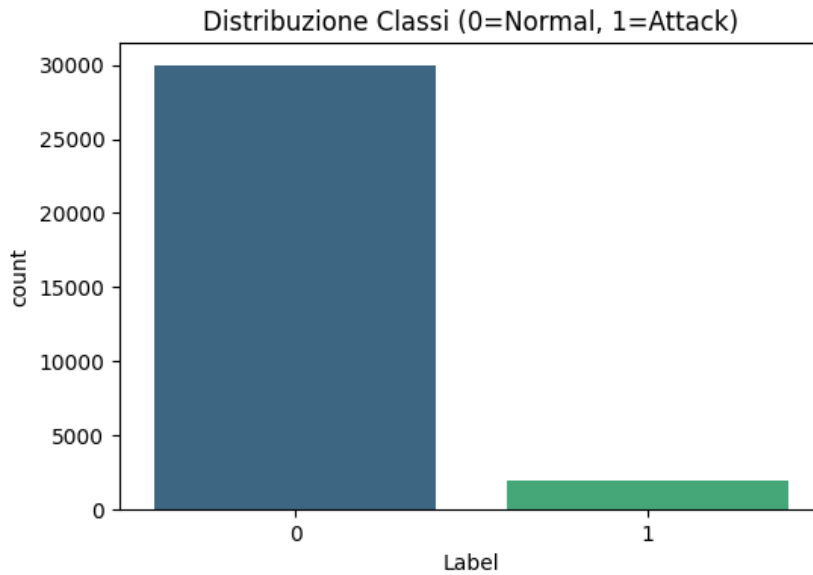


Figure 2: Class Distribution: The dataset is highly imbalanced, necessitating Weighted F1-Score evaluation.

3.3 Sensor Distribution Analysis

To further investigate the discriminative power of individual sensors, we analyzed the distribution of the top features using boxplots, as shown in Figure 3. The plots compare the range and quartiles of sensor readings during *Normal* (0) and *Attack* (1) states. It is observed that for many critical sensors (e.g., FIT201, FIT501), the distributions overlap significantly. This overlap confirms that simple threshold-based detection is insufficient for this domain, justifying the use of complex ensemble models like Random Forest and XGBoost that can capture non-linear correlations between multiple sensors.

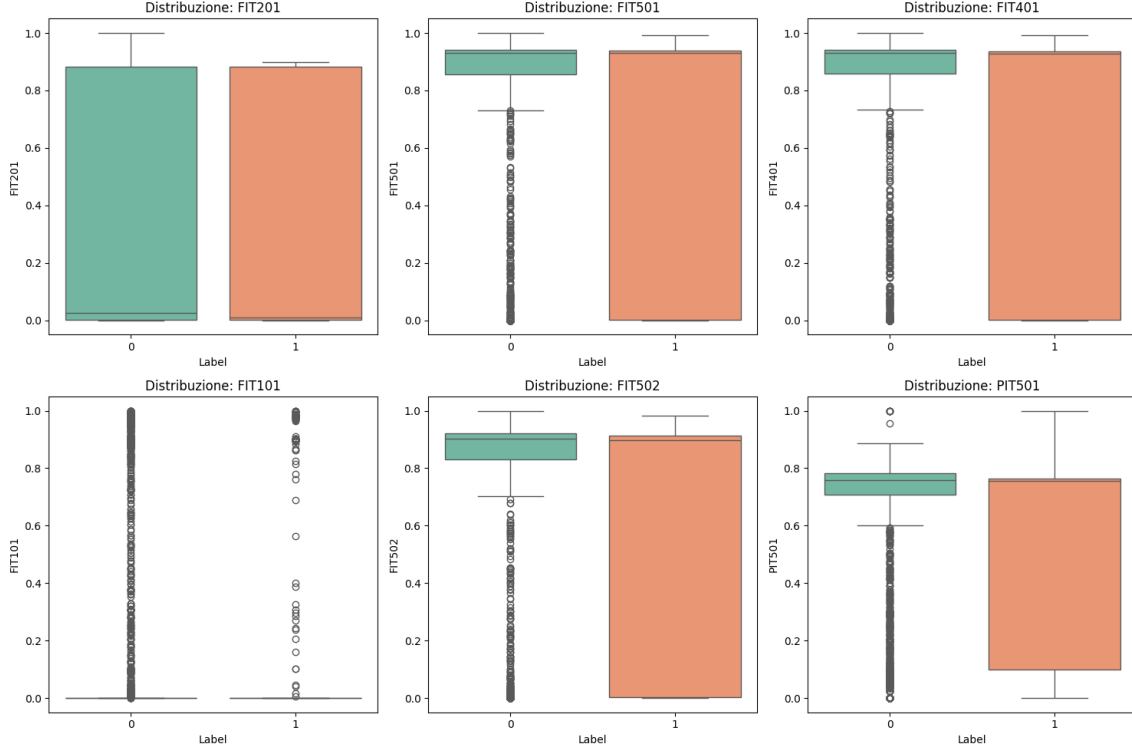


Figure 3: Boxplots of representative sensors. The overlapping distributions between classes highlight the complexity of the detection task.

4 Experimental Results

4.1 Baseline Performance

In normal conditions, ensemble models (RF, XGB) achieved superior performance compared to the single Decision Tree.

Model	F1-Score (Weighted)	Overall Accuracy
Decision Tree	0.884	88.2%
Random Forest	0.903	92.6%
XGBoost	0.907	93.4%

Table 2: Baseline performance on Test Set C.

4.2 Adversarial Impact (All Attacks)

We evaluated the models against four different attack strategies. Table 3 presents the comprehensive results.

Model	Attack	F1-Score (Adv)	Drop	Time (s)	Similarity
Decision Tree	ZOO	0.673	-23%	3.40	0.999
	Boundary	0.193	-78%	8.75	0.963
	HopSkipJump	0.193	-78%	12.27	0.979
	Sign-OPT	0.193	-78%	18.86	0.999
Random Forest	ZOO	0.841	-6%	144.08	0.997
	Boundary	0.286	-68%	133.28	0.921
	HopSkipJump	0.315	-65%	572.48	0.922
	Sign-OPT	0.011	-99%	793.67	0.916
XGBoost	ZOO	0.817	-9%	13.50	0.996
	Boundary	0.850	-6%	7.01	0.992
	HopSkipJump	0.871	-4%	8.07	0.999
	Sign-OPT	0.421	-53%	63.63	0.914

Table 3: Comprehensive attack results. While Random Forest collapses under decision-based attacks (Sign/Bound/Hop), XGBoost maintains high resilience against Boundary and HopSkipJump.

4.3 Visual Analysis: Confusion Matrices

The following figures illustrate two contrasting scenarios: the catastrophic failure of Random Forest under Sign-OPT, and the remarkable resilience of XGBoost under HopSkipJump.

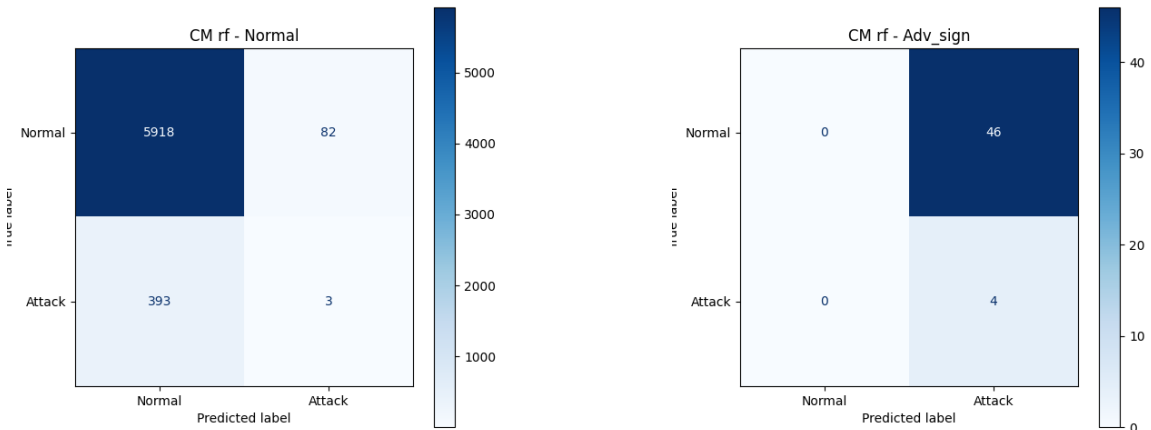
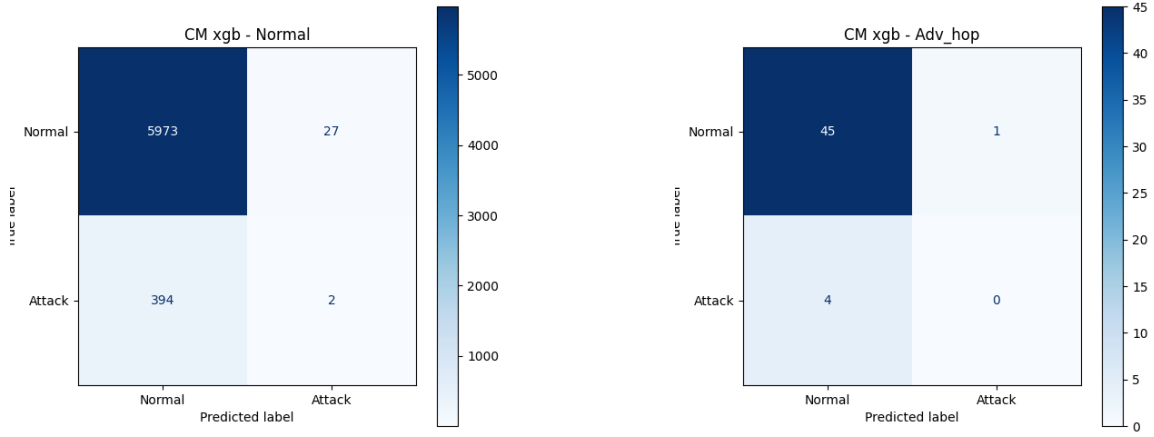


Figure 4: **Random Forest Failure:** Under Sign-OPT attack, the model fails to detect almost all attacks (False Negatives).



(a) XGB: Normal Conditions

(b) XGB: Under HopSkipJump (Robust)

Figure 5: **XGBoost Resilience:** Unlike RF, XGBoost maintains high detection rates even under HopSkipJump attack.

5 Discussion: Efficiency vs. Security

A core contribution of this study is the evaluation of how **Feature Engineering** impacts both the operational efficiency of the IDS and its adversarial vulnerability. We compared the *Processed* dataset (58 features) with the *Refined* dataset (46 features).

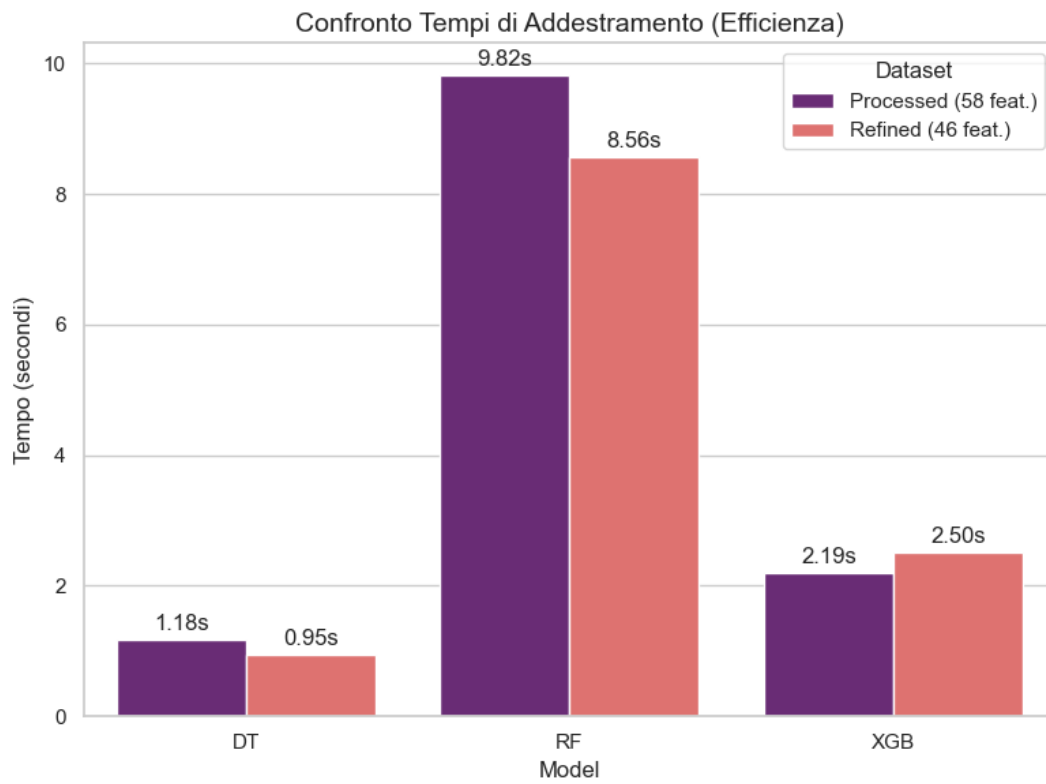
5.1 Impact on Performance and Training Efficiency

As illustrated in Figure 6a, the removal of 12 redundant and highly correlated features did not degrade the models' predictive capabilities. In fact, the Decision Tree (DT) showed a slight improvement in F1-Score, likely due to the reduction of noise in the splitting process.

Furthermore, Figure 6b demonstrates a clear gain in computational efficiency. The training time for the Random Forest model decreased by approximately 13% (from 9.82s to 8.56s). In a real-world deployment on edge devices or embedded systems within a water plant, this reduction translates to lower power consumption and faster model updates.



(a) F1-Score Stability



(b) Training Time Efficiency

Figure 6: Comparison between Processed and Refined datasets. Feature selection maintains performance while improving training speed.

5.2 The Security Paradox

Despite the gains in efficiency, the reduction of the feature space introduced a significant security trade-off. As shown in Figure 7, the time required for the ZOO attack to generate successful adversarial examples for the Decision Tree dropped by 28% (from 4.72s to 3.40s).

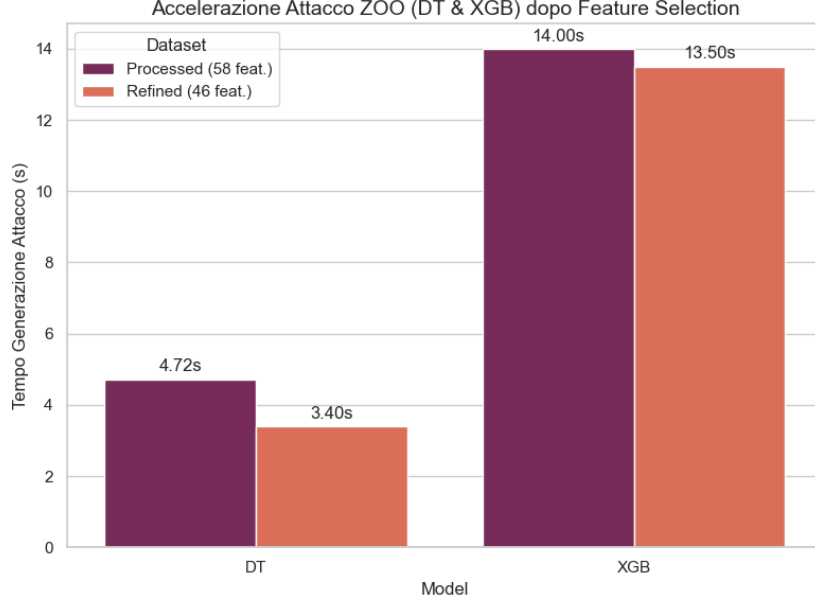


Figure 7: Acceleration of ZOO attack generation after feature selection.

This result reveals a **Security Paradox**: by optimizing the dataset for the defender (removing redundancy), we simultaneously optimized the search space for the attacker. With fewer dimensions to perturb, adversarial optimization algorithms converge significantly faster, reducing the "window of opportunity" for the defender to detect the ongoing attack generation process.

6 Conclusion

This project successfully transferred the adversarial attack methodology from the Automotive to the Water Treatment domain.

- **Sign-OPT** emerged as the most dangerous attack, capable of bypassing Random Forest defenses ($F1 \approx 0$), unlike in the reference thesis where ZOO was the primary focus.
- **XGBoost** proved to be the most balanced model. It showed exceptional resilience against geometric attacks (Boundary/HopSkipJump) where Random Forest failed, although it remains vulnerable to Sign-OPT.
- **Security Trade-off**: Feature engineering improves efficiency but can reduce the computational cost for the attacker.

References

- [1] Vita Santa Barletta and Samuele del Vescovo. “Adversarial Attacks on IDS and Multidomain Impact Analysis for Threat Intelligence in Military Automotive Scenarios”. Reference Thesis. Software Engineering Research Lab, 2024.
- [2] iTrust. *CISS 2020 Dataset: Critical Infrastructure Security Showdown*. https://itrust.sutd.edu.sg/itrust-labs_datasets/dataset_info/. Secure Water Treatment (SWaT) Testbed. 2020.
- [3] Maria-Irina Nicolae et al. “Adversarial Robustness Toolbox v1.0.0”. In: *arXiv preprint arXiv:1807.01069* (2018).